NYC Taxi Mobility

Team: Paola Biocchi 340437 | Siba Smarak Panigrahi 352339
COM-480 Data Visualization · EPFL · Spring 2026
elitehackers-six.vercel.app

900M	10	5
trips processed	years (2015–2024)	linked views

Introduction & Project Goal

New York City (NYC) taxis move millions of people every day, yet the patterns behind that movement are invisible in raw numbers. Our project transforms NYC taxi trip data into an interactive web narrative and a fully linked analytical dashboard.

> ***Core question**: How do taxi usage patterns vary across time and areas, and what underlying factors drive these variations?*

We address this through three analytical lenses simultaneously:

- **Temporal dynamics**: demand patterns from hour-of-day to annual events
- **Trip characteristics**: interactions between trip fare, distance, tip, and duration
- **Behavioural anomalies**: dips and surges in taxi usage tied to external events, such as **COVID-19**, weather, or festivals

The final product serves **first-time users** who want a guided story and also **analysts** who want to explore freely. A **story** mode guides through scroll-triggered story steps, and a free exploration **dashboard** mode exposes the full linked dashboard with global and local features for curious users to explore and understand. Please refer to our screencast for a quick overview of the product's features.

Dataset & Data Pipeline

The NYC TLC Trip Record dataset is distributed as monthly Parquet files. We process three vehicle types with definitions that remain stable throughout: **Yellow Taxis (744M trips)**, **Green Taxis (67M trips)**, and **For-Hire Vehicles (FHV, 98M trips)**. We exclude High-Volume FHV (HVFHV) because before February 2019, HVFHVs were bundled in a single FHV file. Including them would introduce a definitional discontinuity that makes pre- and post-2019 FHV volumes incomparable, so we restrict FHV to records after 1 February 2019 and drop HVFHV entirely. Additionally, this helps keep data storage and processing tractable throughout the project, as HVFHV was by far the largest category (∼3x larger files than yellow taxis). Further, we also exclude 2025, which was incomplete at the time of data processing, retaining only complete calendar years.

We built a Python data processing pipeline in five sequential stages. **Download** automatically fetches monthly parquet files; **Validate** checks columns and detects duplicates; **Preprocess** cleans invalid entries, temporal anomalies (drop-off before pick-up), harmonizes columns across taxi types; **Export** produces CSV files used for the EDA in Milestone-1, and **Aggregate** summarises the cleaned records into the six JSON files loaded by the web. At the end, we have 38 GB of data to work with, available on HuggingFace.

High-level Overview of Visualizations

Based on our motivation and three problematic axes, we decided to build the final set of visualizations (views) on the website around **two distinct experiences**. We design the **story** page as **scrollytelling**: a single chart remains on screen and is redrawn as the reader scrolls. We tell the entire story with the following two arcs.

Arc 1: The city keeps a rhythm

Overlaying the number of trips across all ten years resolves daily noise into recurring structure, such as the **July 4th dip**, the **year-end cliff** (holidays), and the **early-September rise** (end of the summer season).

Arc 2: The rhythm breaks

We then zoom in on the year 2020 and highlight the collapse during **Covid** and the drastic change in the weekly traffic shape due to the citywide pause. Over time, the Arc 1 regularities slowly reappear.

We then designed the **Dashboard** page as a **free-exploration mode**, with **five linked interactive visualizations**, particularly for analysts who want to query the data directly. Below we list our final versions of these visualizations.

`Annotated Event Timeline`

Daily total trips, presented as a line chart with colour-coded event markers, and allows users to explore custom days and their impact on traffic. ***Purpose:*** The entry point to interact with the NYC TLC data; clicking on an existing event marker, or any arbitrary day, depicts before/during/after impact statistics along with an explanation.

`Trip Volume Over Time`

Monthly trips stacked by three taxi types. ***Purpose:*** The visualization shows the dataset's scale and allows users to select a time period and analyze patterns in detail.

`Weekly Pulse Heatmap`

A grid of average demand by day and hour showing the peak demand periods. ***Purpose:*** Expose the city's weekly rhythm and highlight the detailed hour-wise demand.

`NYC Zone Choropleth`

Pickup volume across all 265 taxi zones, highlighting which neighbourhoods have the highest demand and the peak-demand month. ***Purpose:*** Provide a spatial dimension to trip characteristics in the form of a demand overlay on a map.

> `Trip Anatomy Explorer`
>
> A scatter plot of trips with metadata on axes and showing the statistical relationship between them. ***Purpose:*** Study the "trip characteristics" axis of our problematic.

Along with the individual local features for each visualization, we keep **global controls** (Figure 1) at the beginning of the dashboard, including a taxi-type toggle, a year-range slider (between 2015–2024), a borough (area) selector, and a **reset** button to clear filters. These can filter `Weekly Pulse Heatmap`, `NYC Zone Choropleth`, and `Trip Anatomy Explorer` views. We added the year-range slider and the area selector, different from Milestone 2 (where we intended to add them within views), to keep the design intuitive.

Figure 1: Global controls: taxi-type toggles, year-range slider, area selector, and reset.

Implementation Details

After deciding on the visualizations, we started by implementing the website. We shared the **first website version**, with primarily static placeholders and minimal interactivity, within Milestone 2, and then developed the **full version** in Milestone 3 (tested in Google Chrome). We first created **preliminary hand-drawn sketches** (shared in Milestone 2) and then implemented and optimized the visualizations.

Main Tech Stack

D3.js 7
Used for all rendering, scales, and interaction. Each view exports an `init(container, data)` factory.

Vite 6
Modern build tool for the dev server and production build. Fast live reload sped up development.

Scrollama 3
Drives the scroll-step narrative, triggering each view on the landing page as the reader scrolls.

TopoJSON
Encodes the taxi-zone map geometry compactly by sharing arcs between adjacent zones.

Python
Data pipeline and **aggregation**: the pipeline converts raw Parquet files into compact JSON aggregations, precomputed for all the views offline, so that each view loads only a handful of small JSON files and avoids delays when rendering.

Vercel
Handles **hosting** for our static web application. Each push triggers an automatic build and gives us a permanently available site with zero server maintenance. The code itself is ES-module JavaScript with **no UI framework**; each view is a self-contained factory function, which keeps the deployed bundle small and fast.

Good practices we followed : We created a **central filter bus** for our dashboard, which is a tiny publish/subscribe store and the global controls, `Annotated Event Timeline` and

`Trip Volume Over Time`, can *publish*, while all views *subscribe*. This single source of truth makes the views "linked". Next, we **pre-compute and aggregate** 900M raw trips into JSON files, so the browser never avoids delays when rendering updates from the central filter bus. Finally, all **statistics on the webpage are recomputed** with any event triggers, and we retain **secrets on the server side** without logging them to the browser.

Main Visualizations: From Initial Concepts to Final Design

Under Milestone 1, we first built a separate **EDA dashboard**. It covered a **high-level overview of the data**, including day-of-week patterns, fare and tip by hour, a pickup choropleth, and a borough origin-destination matrix. We followed these observations to first make the **hand-drawn sketch** (Milestone 2), then the scrollytelling **story** on the landing page, and **dashboard** with five linked visualizations.

Scrollytelling

We wanted to create an **informative** and **attractive main landing page** for new users, to show them the capabilities of data visualization on NYC trip data. For this, we built a concrete story grounded in the dataset. One evident theme was the significant decrease in trip volume due to **Covid**, which immediately prompted us to **overlay the distribution of trip volume across all years**. The plot showed us **regularities across years** that coincided with the **events associated with this time period** (see Figure 2). However, we observed that this behaviour was less evident in 2020, and in subsequent years the pattern gradually reverted.

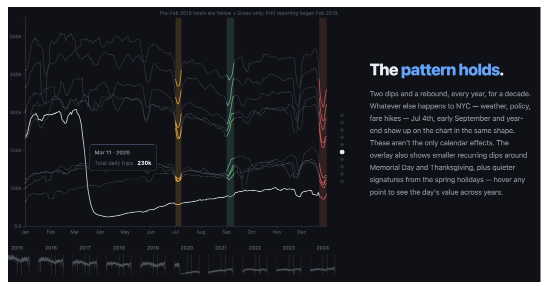

Figure 2: The yearly pattern overlay. Observe the similar behaviour around **July 4th**, **end of year**, and **early September**. The highlighted white line doesn't adhere to this.

As a consequence, we plotted additional patterns for before, during, and after **Covid**, in particular, weekly trip volumes (see Figure 3). We identified a stark contrast in usage during this disruptive period, which was also fueled by the city-wide pause order. In conclusion, we developed this **story** to show that there are underlying patterns in the vast fluctuating data that can be disrupted by a once-in-a-lifetime event, thereby urging users to visit the dashboard and explore more such patterns. Next, we describe each individual **dashboard** visualization in more detail.

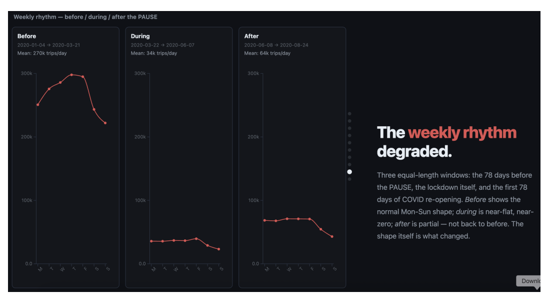

Figure 3: Distinct weekly rhythm before, during, and after **Covid** pause. The peak demand during midweek before the pandemic dropped to a small, constant level throughout the week.

Annotated Event Timeline

This view is our "main" interactive visualization consisting of **a line plot with event markers**, where **hovering** over the line plot shows the date and the total number of trips. It encompasses the **temporal dynamics** and **behavioural anomalies** axes. It allows users to: **A)** analyze predefined events and **B)** probe other patterns to understand traffic behaviour. According to our initial idea, presented in Milestone 2, we included a few **major events**

in the visualization, such as COVID reopening and weather-related (blizzards, storms, etc.), as shown in Figure 6. Generally, these events have a disruptive impact (e.g., a **26% decrease** in average trips during Winter Storm Kenan, Jan 2022). Each event is also associated with a **quantification**, in which we compute the average and variance (volatility) of the number of trips and compare with baseline trends.

Figure 4: Storm Kenan in Jan 2022.

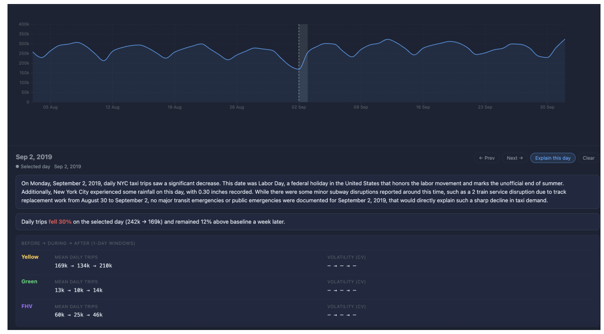

Figure 5: "Explain this day" feature .

That said, we observed a significant number of days that reflected meaningful changes on the plot (e.g., Half Sun Manhattanhenge on May 29, 2023, Red Bull Music Festival on May 30, 2015, other holidays like Labour Day). Therefore, we wanted to expose them to users to make this container as general as possible. Thus, we designed the system such that **clicking on the plot** would observe local trip distribution around the day and open an "**Explain this day**" button powered by an LLM (Gemini-2.5-Flash-lite). **Clicking** this button prompts the LLM to perform a web search and provide a potential reason for the impact. This design also avoids overcrowding with many events and information and goes beyond the Milestone 2's deliverables. This is shown in Figure 5.

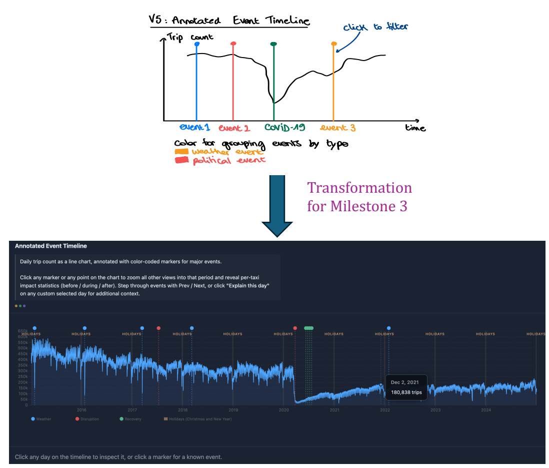

Figure 6: Transformation of the M2 sketch into the final visualization.

Trip Volume Over Time

This view is similar to `Annotated Event Timeline` and shows **stacked monthly trips** per taxi type. We have kept this similar to our proposal in Milestone 2. The primary goal is to show the dataset's scale, global trends, and long-term structural changes, as shown in Figure 7. It has a **brushing feature** that selects a specific time period to closely examine changes across all other visualizations. This container, along with the year-range slider, allows us to **look more deeply** into the `Annotated Event Timeline` and select interesting days for further analysis. For example, if a user is interested in observing the effect of the Manhattanhenge event in 2023, which is generally in May every year, brushing a time period between April and June of a year will show the dips and rises in `Annotated Event Timeline`, and the user can then select a particular day to investigate more (shown in Figure 8). Finally, **hovering** in the normal state (no brushing operation) shows the monthly trips for each taxi type.

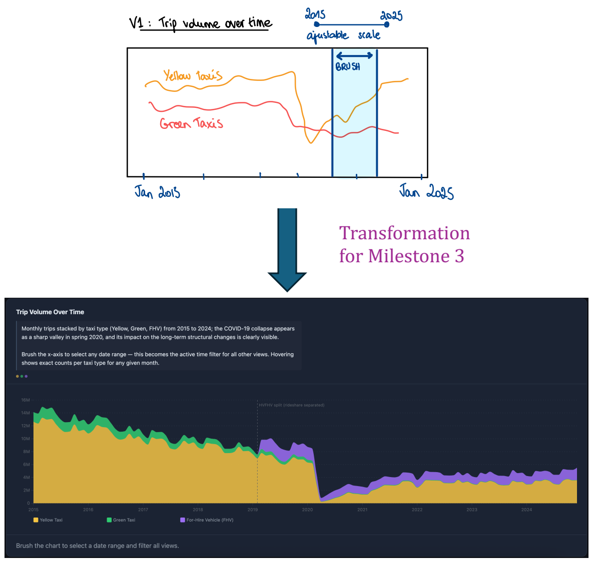

Figure 7: M2 sketch to visualization.

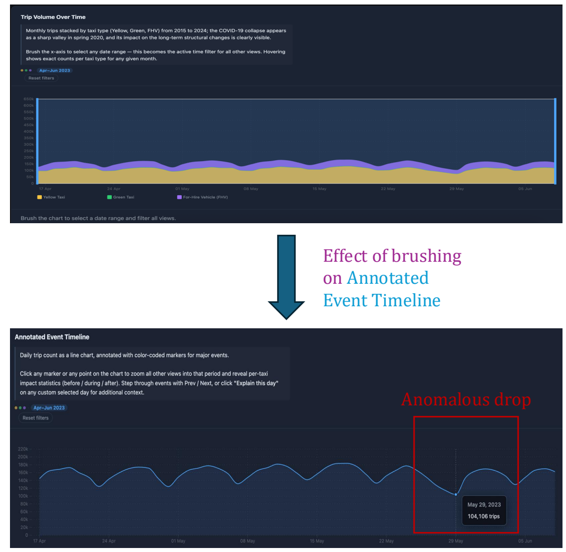

Figure 8: Annotating an interesting day.

Weekly Pulse Heatmap

While the above visualizations focus on temporal and behavioural aspects, the remaining visualizations focus on the **trip characteristics** problematic axes. We closely follow the description of this visualization as per Milestone 2. The main goal is to **highlight day- and hour-level variations** and to show the trends for each taxi type. In particular, it shows the average trip demand by hour of the day and day of the week in a 7×24 (week×hour) **grid heatmap**, and **hovering** over each grid cell displays the exact day, hour, and average number of trips. Furthermore, **clicking a row** opens the full 24-hour demand profile for that day, with one line per taxi type. Finally, we added a **normalize columns** toggle (beyond the deliverables in Milestone 2) to see trends across the days of the week for each hour. A snapshot is shared in Figure 9.

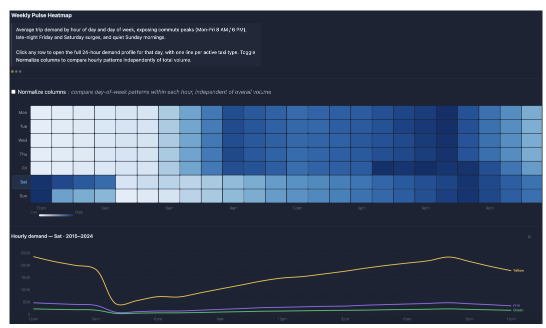

Figure 9: The weekly pulse heatmap.

As expected, we observe **commute peaks on weekdays between 9-10 am** and **6-7 pm**, which align with usual office hours, and the evening peak demand shifts to 8-9 pm on weekends (perhaps dinner). Next, nightlife is obvious, with **high late-night demand on Friday and Saturday** (midnight - 1 am). Interestingly, we observe an **early-evening peak on Friday from 4-6 pm**, indicating that people leave their workplaces earlier. Beyond these observations, when we activate the normalize toggle, we find that **most very late-night travel** (between 1 am - 5 am) occurs **primarily on Saturday**, followed by Sunday, which further supports the weekend life. These trends are uniform across taxi types, and with global controls, these trends can be discovered for specific time periods.

NYC Zone Choropleth

This visualization adds a spatial dimension to the trip characteristics by presenting pickup volumes across NYC's 265 zones as a **choropleth map**, with volume represented as a **heatmap on the predefined NYC map with boroughs and neighbourhoods** (created with TopoJSON). The darker shades of red indicate higher pickup counts. Inspired by our preliminary EDA in Milestone 1, where we identified travel between the zones, we keep the same six boroughs (or areas): EWR, Staten Island, Manhattan, Bronx, Queens, and Brooklyn. We included the **hover** feature, which **details** the number of pickups and the neighbourhood within the borough **in a separate dialogue box**. Additionally, **clicking on the neighbourhood** (within the boroughs) shows the distribution of pickups across taxi types and the peak demand month, as shown in Figure 10.

Now, from this visualization, we first observe that **Manhattan accounts for 73% of all 900 million pickups**, with the Upper East Side and Midtown Center neighbourhoods as hotspots. All the major neighbourhood hubs have **January as the peak demand month**. Beyond these, the airport hubs have higher pickups, such as **JFK** and **LaGuardia**. **EWR** (Newark airport) is relatively quiet, primarily because JFK and LaGuardia are more prominent than EWR for international and domestic travel. This identification of major hubs would not be possible without processing and adding the spatial information. Using global controls, these insights can be further found for specific time periods.

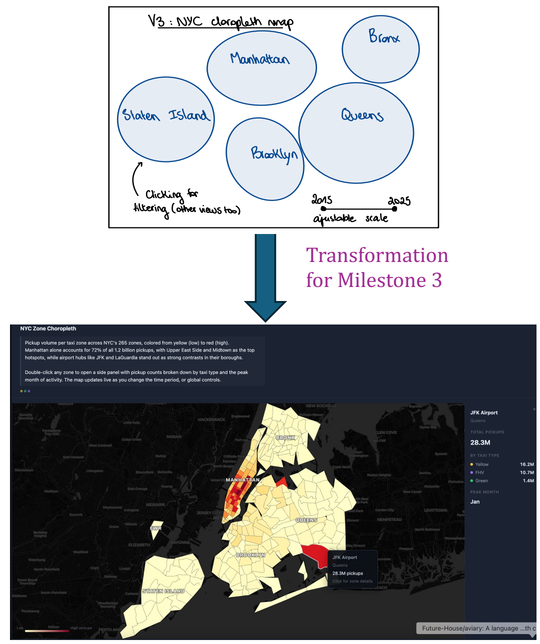

Figure 10: The pickup demands overlaid on the NYC zone geographical map.

Trip Anatomy Explorer

This visualization shows relationships among trip metadata variables: distance, fare, tip percentage, and duration. We ignored passenger information due to its higher discreteness. We built on the Milestone 2 proposal to represent **each variable on a single axis** and create **a scatter plot** with trips as points, **coloured by the hour** of the day. In addition, we always compute and show the strength of the linear relationship among these variables using an R^2 value and a **best-fit dotted line**, which indicates how well the x-variable explains the variance in the y-variable. Consequently, we subsampled 50,000 points uniformly across all months to avoid predictive modelling on large matrices. This visualization is shown in Figure 11. There was a lack of metadata for FHV trips; thus, we retain only yellow and green taxis in this view. Finally, **hovering** over the scattered points displays the metadata for that trip and highlights all other trips from the same hour of the day.

Our visualization provides insightful trip characteristics. First, **most variables** (e.g., distance vs. fare, distance vs. duration) exhibit **strong linear correlation** with $R^2 > 0.9$. However, **none of the variables are correlated with the tip percentage**. An interesting observation emerges between distance and fare: on average, for the same distance, non-peak hours generally result in lower fares than peak hours. Similarly, between distance and duration, we observe that similar distances take shorter duration in non-peak hours and lie beneath the best-fit line (Figure 11). These conclu-

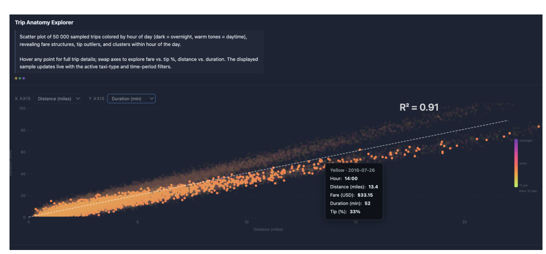

Figure 11: Explore the relationship between different trip characteristics.

sions are only possible because of our design choice to colour trips by hour of the day.

In addition to the local features described above, one of the most crucial components for insight discovery is combining these features with the temporal axes (via the year-range slider or brushing in `Trip Volume Over Time`, or selecting arbitrary dates in `Annotated Event Timeline`). We demonstrate this in the landing page scrollytelling **story**. Therefore, **dashboard** and **story** combined successfully cover all aspects of our problematic.

Challenges

Beyond our path to final visualizations, we highlight a few dataset- and feature-specific interesting challenges and how we handled them. First, the **dataset scale**. The processed data is 38GB spread across hundreds of monthly Parquet files. Live computation in the browser is not viable, so we built an **offline pipeline** that **pre-computes JSON aggregations**, allowing the site to use them and update instantly on any query. Second, in Milestone 2, every visualization cross-filtered with every other, which sometimes resulted in sparser views. We redesigned it to separate into **global** and **local** filters, handling only meaningful interactions. Finally, the LLM-based "Explain this day" was initially prone to **hallucinations**. As a result, we grounded the LLM using a web search tool and provided the change in trip demand for the requested days as input to avoid hallucinations.

Peer Assessment

The two of us worked closely throughout, including milestone write-ups, the process book, the screencast and final implementation debugging. Visual sketches and designs were discussed during brainstorming sessions (in-person on campus) with equal contribution from both members. Implementation responsibilities were split equally as follows:

Member	Primary contributions
Paola Biocchi	Exploratory data analysis (EDA) for Milestone-1; Python scripts for JSON aggregation; `Trip Volume Over Time` stacked area chart; `Weekly Pulse Heatmap`; `NYC Zone Choropleth`; `Trip Anatomy Explorer`; global controls
Siba Smarak Panigrahi	Data download and processing pipeline (validation, cleaning, feature engineering); Scrollytelling **story** narrative, the central filter bus architecture; `Annotated Event Timeline`; "Explain this day"; vercel deployment; global controls

In our project, all idea development (including the **story** and **dashboard**) and analysis were carried out solely by the team members. However, LLMs assisted with code development for debugging (e.g., interpreting error messages), implementing repetitive structures, and improving code readability. All **AI-assisted code** was **reviewed, understood, and critically tested** by the team, and adapted where necessary before inclusion, ensuring we retained a solid grasp of the underlying JavaScript. Finally, we used ElevenLabs to generate the voiceover for our screencast, using a **written script we prepared**.

Conclusion

We hope our current insights and website will help people decide how to optimize their taxi trips in NYC and prepare for different taxi traffic patterns!